

ASHUTOSH SRIVASTAVA

B.Tech CSE | AI & Analytics | Full-Stack & AI/ML Engineering

📍 Pune, Maharashtra, India | 📞 +91 92196 56385 | ✉️ ashutoshsrivastavasoc19@gmail.com | 🔗 LinkedIn | 📄 GitHub | 📖 LeetCode

PROFESSIONAL SUMMARY

Third-year B.Tech Computer Science student specializing in Artificial Intelligence & Analytics, building production-grade full-stack and AI systems — hybrid retrieval and RAG platforms, real-time telemetry and network intelligence, and multi-role workflow platforms. Comfortable across Python/FastAPI backends, React and Next.js frontends, PostgreSQL/pgvector data layers, and applied machine learning, with a strong bias toward measurable, well-tested engineering.

WORK EXPERIENCE

Full Stack Developer Intern | Internship

June 2026

Clyfar Solutions

On-site • Nashik, India

- Engineered responsive web application interfaces with JavaScript, HTML5 and CSS frameworks, delivering consistent rendering across desktop and mobile breakpoints.
- Partnered with UI/UX designers and contributed to peer code reviews, improving component reusability and maintainability across the frontend codebase.

Technologies / Skills Used : JavaScript, HTML5, CSS3, Responsive Design, Git

AI-eCommerce Assistant Intern | Internship

December 2024 – January 2025

iNeuron.AI

Remote • India

- Built backend and frontend proof-of-concept modules for an AI eCommerce Assistant, validating assisted product-discovery workflows end to end.
- Integrated AI-powered modules into core e-commerce flows, enabling automated assistance across catalogue browsing and customer interaction paths.

Technologies / Skills Used : Python, AI Module Integration, REST APIs, Frontend Development

Web Development Intern | Internship

March 2024 – April 2024

Oasis Infobyte

Remote • India

- Designed and developed web application features from requirement to delivery, translating product specifications into functional, reviewed interfaces.
- Diagnosed and resolved frontend defects across components, hardening stability and consistency of shipped features.

Technologies / Skills Used : HTML5, CSS3, JavaScript, Git, Debugging

EDUCATION

MIT ADT University | Pune, India

2024 – 2028

B.Tech, Computer Science & Engineering — Artificial Intelligence & Analytics

CGPA : 8.0 / 10

School of Computing • Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, Software Engineering

St. Dominic Savio College — Class XII, Science with Mathematics, Lucknow

CBSE

City Montessori School — Class X, Lucknow

92%

PROJECTS

AEGIS — Autonomous Enterprise Intelligence & Decision Operating System

GitHub

- Architected an enterprise decision operating system closing the sense → predict → reason → decide → act → learn loop across data, intelligence, AI, decision and action platforms.
- Engineered a Bronze/Silver/Gold data platform with schema registry, data-quality validation, health scores, quarantine and catalog lineage over a Kafka-compatible streaming backbone with consumer groups, partition management, lag monitoring, idempotency, dead-letter queues and replay.
- Implemented RAG retrieval, agentic reasoning and LLM routing with MCDA scenario simulation and governed action execution — sagas, postconditions and outcome attribution — under multi-tenant security.

Technologies / Tools Used : Python, Event Streaming (Kafka-compatible), RAG, LLM Orchestration, AI Agents, Vector Search, Data Engineering, Multi-Tenant Architecture

NOVANET — AI-Native Network Intelligence & Discrete-Event Simulation Platform

GitHub

- Built an autonomous decision engine performing predictive adaptive routing and in-flight failure recovery, driven by linear-regression and EWMA congestion forecasting (MAE 0.000045, RMSE 0.000064, R^2 0.7987).
- Engineered snapshot-first telemetry from real system metrics via psutil, exposed through FastAPI REST endpoints and WebSocket streams into a real-time React observability dashboard.
- Validated the platform with 224 automated tests and scale benchmarks from N=10 to N=10,000 nodes, including documented 5,000-node graph configurations and resource-limit handling.

Technologies / Tools Used : Python, FastAPI, WebSockets, React, Machine Learning, psutil, Discrete-Event Simulation

VERIDEX — Enterprise Knowledge Intelligence Platform

GitHub

- Engineered a hybrid retrieval pipeline fusing pgvector HNSW dense search with PostgreSQL tsvector/GIN keyword search through Reciprocal Rank Fusion and cross-encoder reranking, sharpening passage relevance.
- Implemented grounded LLM generation with server-side citation verification, SSE streaming and controlled abstention, preventing answers unsupported by source documents.
- Delivered multi-tenant workspace isolation, HttpOnly session security and versioned PDF/DOCX/TXT/Markdown/CSV ingestion for auditable document intelligence.

Technologies / Tools Used : Next.js 14, React 18, TypeScript, Tailwind CSS, FastAPI, Python 3.11, Pydantic v2, SQLAlchemy 2 (async), Alembic, PostgreSQL, pgvector, Playwright

PHARMALOOP — Pharmaceutical Reverse Supply Chain Platform

GitHub

- Designed a four-portal reverse supply chain system — medical shop owner, pharma company, logistics partner and super admin — digitising the expired-medicine return workflow across stakeholders.
- Implemented OTP authentication, real-time return, refund and location tracking, medicine management and Excel export over database-backed workflows, replacing manual reconciliation.

Technologies / Tools Used : Next.js, React, TypeScript, Tailwind CSS, Node.js, Express, MongoDB, Socket.io

SKILLS

Programming Languages : Python, C++, Java, JavaScript, TypeScript, SQL

Frontend : React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & APIs : FastAPI, Node.js, Express.js, REST APIs, WebSockets, Server-Sent Events, SQLAlchemy

AI / Machine Learning : Machine Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, NLP

Data & Search : PostgreSQL, MongoDB, pgvector, Vector Search, Hybrid & Semantic Search, Data Engineering, Event Streaming

Engineering & Tools : Git, GitHub, Docker, Linux, Playwright, System Architecture, Real-Time Systems, Multi-Tenancy

ACHIEVEMENTS & OPEN SOURCE

- GeeksforGeeks Student Chapter** — Tech Team Member: built chapter technical platforms and orchestrated technical setups for campus events, delivering reliable execution across sessions.
- Open Source Connect India** — Contributed bug fixes to open-source online-education tooling and mentored new contributors through their first accepted pull requests.
- Algorithmic Problem Solving** — Sustained LeetCode practice across data structures, algorithms, dynamic programming, binary search and graph patterns.

CERTIFICATIONS & LANGUAGES

Certifications : Machine Learning with Python — IBM | Introduction to Generative AI — Google Cloud | Python Essentials 1 & 2 — Cisco | AI for All — Intel

Languages : English, Hindi, Marathi